

Supplementary material for “Walls or cascade? A pre-registered test of wall dissipation in decaying magnetohydrodynamic turbulence”

Alessandro De Rosi^{a)}

The University of Manchester, Manchester M13 9PL, United Kingdom

(Dated: 3 October 2026)

Equations, sections, figures and tables numbered without an S are those of the paper. Every number below that comes from a run is printed, with its source, by the script that produces the paper’s numbers (`paper/tg_mhd/numbers.py` in the repository).

S1. VERIFICATION OF THE METHOD

The parity wall. Three tests in the repository check Eq. (6) (`validation/mhd_parity_wall.cpp`). First, with free slip and either parity, the box of N nodes reproduces the periodic 2π box of $2(N-1)$ nodes to 10^{-14} , node for node and transient included. Second, $\mathbf{B} \cdot \mathbf{n}$ on the wall nodes stays at 10^{-32} . Third, in the no-slip box the exact decay of the conducting-wall eigenmode $\mathbf{B} = B_0(\sin x \cos y, -\cos x \sin y, 0)$, which leaves the fluid at rest, converges at second order (order 2.00 over $N = 17, 33, 65$).

The two implementations. The C++20/Kokkos code and the CUDA code share no source. Their host builds agree to every printed digit over 815 steps at $N = 65$, $\text{Re} = 200$, with free and no slip, and the CUDA build on the device agrees with the Kokkos reference to within 5.5×10^{-16} of each diagnostic’s scale, in six verification jobs, each with free and no slip, the last also with the smooth start and the energy-matched box of Sec. S8. The runs behind the paper’s numbers took 34.1 GPU-hours: the ladder, its two controls, and a second round with the top rung, the insulating pair, and reruns of the lower rungs that also write the viscous and Ohmic parts of the profile. The reruns that wrote the fields of Sec. IV F took 1.5 more, those that wrote the energy budget of Sec. IV G 15.1, and the first ladder, run with the estimator of Sec. S3, 8.2.

Run C2 of Pouquet et al. The free-slip box A at $\text{Re} = 1000$ is run C2 of Ref. 1. Its minimum ratio of magnetic to kinetic energy is $E_M/E_V = 0.3695$ and 0.3693 at $N = 384$ and 512 , reached at $t = 7.3$, against their 0.35 , and the first maximum of Ω_M/Ω_V after its initial fall is 2.154 and 2.159 , against their “2.”. Our ratio is converged to four figures between the two grids and lies 5.5% above theirs, which is quoted to two figures.

We attribute the gap to the resolution of C2. It is the coarsest run of Ref. 1: pseudo-spectral on an equivalent grid of 128^3 over the period 2π , i.e. 64 points across our

box. Its Table 1 lists $\delta k_{\text{max}} = 0.8$, where δ is the logarithmic decrement of a fit $E_T(k) \propto k^{-n} e^{-2\delta k}$ to the energy spectrum and $k_{\text{max}} = N/3$. That is below $\delta k_{\text{max}} \approx 1$, at which the authors stop an ideal computation, and below the $\delta k_{\text{max}} \geq 2$ by which they call their 2048^3 runs well resolved; their other two 128^3 runs have 1.9 and 2.1, and C2 has no companion at higher resolution. Under-resolution moves our own ratio the same way: 0.357 at $N = 97$, reached at $t = 9.8$, and 0.329 at $N = 65$, still falling at $t = 10$. The two codes under-resolve by different mechanisms, so the common direction is suggestive rather than conclusive.

S2. THE RECORD OF THE PROTOCOL

The two outcomes of Sec. I were stated, qualitatively, on 25 September 2026 in a plan that is not in the repository, and nothing there dates them. The rules of Sec. IID were committed with the job scripts. A commit’s date is written by its author, so Table S1 also gives the time at which each commit reached GitHub, from GitHub’s server-side record of the pushes to the repository. A copy of that record, as GitHub’s public interface returned it on 2 October 2026, is kept in the repository (`paper/tg_mhd/provenance/`) with the request that fetched it; GitHub returns it to anyone for about three months after a push. Each rule was on GitHub within 80s of being committed. The estimator was corrected after the first ladder had failed its gate (Sec. S3); the observable it estimates and the gate did not change. The Mach control, and the decision rule and the precision control for the top rung, were fixed after the lower rungs had been run, and seen, and before the runs they judge. Before 30 September the ladder’s script asked only that B–A exceed its band and trend monotonically over at least three rungs; the decision rule of Sec. IID, which is the one applied in the paper, replaced that with a test at the top rung, fixed with the job script that ran it. The second test, of the energy budget (Sec. IV G), was fixed with the commit that also carries the code that writes the budget, so the runs it governs could not have been made with any earlier commit; that commit reached GitHub two seconds after it was made (Table S1).

S3. THE ESTIMATOR OF THE WALL SHARE

Counting whole node layers, every node with $kh < m\delta$, measures a layer of thickness $(\lceil m\delta/h \rceil - \frac{1}{2})h$ rather than

^{a)}Electronic mail: alessandro.derosis@manchester.ac.uk

TABLE S1. When each rule was fixed: the commit, its time by the author’s clock, and the time at which it reached GitHub by GitHub’s push record, on the same day. All times UTC, 2026.

rule	commit	committed (UTC)	pushed (UTC)	lag (s)
observable, peak rule, gate	663929c	27 Sep, 11:57:38	11:58:24	46
estimator of f_m corrected	9f65706	28 Sep, 11:55:32	11:56:52	80
Mach control	cd07f45	29 Sep, 06:22:12	06:22:15	3
decision rule, FP64 control (Re = 2000)	d061ee8	30 Sep, 20:03:29	20:03:34	5
the budget test, ϕ	8376e66	02 Oct, 12:00:24	12:00:26	2
the four controls	f7f55ba	03 Oct, 04:19:57	04:19:59	2

$m\delta$, because the nodes with $d = kh$ fill exactly the slab between $(k - \frac{1}{2})h$ and $(k + \frac{1}{2})h$ (Sec. II): 0.91δ at $N = 384$ and 1.07δ at $N = 512$ for $\text{Re} = 1000$. Our first ladder counted layers. At every rung of every box with a turbulent peak, eight in all, it failed the gate: the worst of f_1 , f_2 and f_4 differed between the two grids by 6.5–24.6%, while ε at the peak agreed to 1%. Moved to the exact band, by interpolating each grid’s three samples, the same series pass, the worst difference becoming 2.9%. Swept across $N = 33$ to 49 at fixed physics ($\text{Re} = 100$), the count makes f_4 jump between 0.52 and 0.60 each time $4\delta/h$ crosses a node, while Eq. (7) keeps it within 0.561–0.566. Any layer, shell or threshold observable on a lattice can do this.

S4. RESOLUTION GATE AND CONTROLS

Table S2 gives the resolution gate and the controls of Sec. III rung by rung.

S5. THE THICKNESS OF THE LAYER

Table S3 repeats the readings of Table III for the layers 2δ and 4δ thick. The bands are the two grid differences of each layer’s share, added, as for δ ; the largest is 0.0056 at the peak and 0.0016 over a time window. The thicker layers read as the thinnest does. B–A is positive in every reading at every rung, from 0.019 to 0.115. From $\text{Re} = 250$ to 2000 it changes by +2% within 2δ and –20% within 4δ over $t = 2$ –6, against –3% within δ ; falls by 12–42% and 23–35% over the windows that reach further into the decay, against 18–43%; and falls by 68% and 67% at the peak, against 51%.

S6. INSULATING WALLS

The pair A’, B’ was meant to show whether the result needs a conducting wall. It cannot. A’ passes the gate (worst 0.8%), and 88% of the dissipation within δ of its walls is Ohmic ($F_1 = 0.31$ and 0.27 at $\text{Re} = 500$ and 1000). B’ has no turbulent peak by $t = 10$ at either Re . It puts more of its dissipation within δ of the walls

than any other box, $F_1 = 0.63$ and 0.69, and these agree between its two grids to 1.4% and 1.7%. So B’–A’ (+0.32 and +0.42 over $t = 2$ –10) compares a box that has not become turbulent with one that has, and under a different initial field from B–A.

B’ is also the only box in which the field crosses a no-slip wall, the configuration of a Hartmann layer.^{2,3} With the field normal to the wall, B_n , and a tangential velocity U outside the layer, the tangential momentum and induction equations in the layer are, in Alfvén units, $\nu u'' + B_n b' = 0$ and $B_n u' + \eta b'' = 0$, primes denoting derivatives along the normal. They give $u = U(1 - e^{-z/\delta_H})$ with $\delta_H = \sqrt{\nu\eta}/|B_n|$, and a dissipation per unit area of

$$\int_0^\infty (\nu u'^2 + \eta b'^2) dz = U^2 |B_n| \sqrt{\nu/\eta}, \quad (\text{S1})$$

half of it viscous and half Ohmic. At $\text{Pm} = 1$, $\delta_H = 1/(\text{Re}|B_n|)$ has the scaling of Kato’s layer, and Eq. (S1) does not depend on Re . The field $\mathbf{B}_I$ of Eq. (5) is normal to every face at $t = 0$, with $|B_n|$ up to $2b_0 = 1.15$ on the two faces normal to z and up to b_0 on the other four. There a Hartmann layer would be $0.14h$ and $0.28h$ thick at $\text{Re} = 1000$, and $0.21h$ and $0.42h$ at $\text{Re} = 500$, on the finer grids. These are estimates from the initial field: we have not measured the layers of B’, and its integrated share agrees between grids that could not resolve them. We draw no conclusion from B’ or from B’–A’.

S7. THE ENERGY BUDGET

The terms. Per probe and per wall-distance layer, out to 8δ , the drivers write twelve terms, each as the layer’s contribution to the box mean: the energies $e_K = |\mathbf{v}|^2/2$ and $e_M = |\mathbf{B}|^2/2$; the viscous and Ohmic dissipation $\nu|\boldsymbol{\omega}|^2$ and $\eta|\mathbf{j}|^2$; the stretching $S = B_i B_j \partial_j v_i$ and the Lorentz work $\mathbf{v} \cdot (\mathbf{j} \times \mathbf{B})$; the advections $-\mathbf{v} \cdot \nabla e_K$ and $-\mathbf{v} \cdot \nabla e_M$; the pressure work $-\mathbf{v} \cdot \nabla p$; the viscous and resistive transports $\nu \nabla \cdot (\mathbf{v} \times \boldsymbol{\omega})$ and $\eta \nabla \cdot (\mathbf{B} \times \mathbf{j})$; and the compressible remainder of the induction, $-|\mathbf{B}|^2 \nabla \cdot \mathbf{v} + (\mathbf{v} \cdot \mathbf{B}) \nabla \cdot \mathbf{B}$. The transports need second derivatives, closed at each wall by the extension of Sec. II.

Checks. The two implementations write the budget identically, to every printed digit after $t = 0$, at $N = 65$

TABLE S2. Resolution gate and controls. Relative differences of f_1 , f_2 and f_4 at the turbulent peak between the two grids of each rung, the band of B–A (the two grid differences of f_1 , added), and the shift of B–A in the control run: Mach, u_0 halved; FP64, double precision at $N = 512$.

Re	N	A: $\Delta f_1/\Delta f_2/\Delta f_4$ (%)	B: $\Delta f_1/\Delta f_2/\Delta f_4$ (%)	band	control shift
250	192, 256	0.7/0.2/0.1	1.0/0.7/0.2	0.0020	–
500	256, 384	0.8/0.5/1.0	2.1/1.5/0.7	0.0030	Mach, -0.0014
1000	384, 512	0.7/0.7/0.3	0.2/0.1/0.2	0.0004	Mach, $< 10^{-4}$
2000	512, 640	3.0/1.3/0.8	0.5/0.4/0.3	0.0021	FP64, $< 10^{-4}$

TABLE S3. B–A in the share of the dissipation within $m\delta$ of the walls, at the turbulent peak and integrated over the time windows of Table III, on the finer grid of each rung. The last column is the change from Re = 250 to 2000.

reading	m	250	500	1000	2000	change (%)
at the peak (pre-registered)	1	0.073	0.065	0.063	0.036	–51
	2	0.075	0.076	0.065	0.024	–68
	4	0.058	0.050	0.064	0.019	–67
integrated, $t = 2-6$	1	0.049	0.048	0.051	0.048	–3
	2	0.047	0.044	0.051	0.048	+2
	4	0.060	0.049	0.050	0.048	–20
integrated, $t = 2-10$	1	0.059	0.055	0.054	0.048	–18
	2	0.055	0.056	0.057	0.048	–12
	4	0.063	0.061	0.059	0.049	–23
integrated, $t = 3-10$	1	0.066	0.059	0.055	0.046	–29
	2	0.066	0.062	0.058	0.044	–34
	4	0.066	0.070	0.063	0.044	–32
integrated, $t = 4-10$	1	0.082	0.072	0.064	0.050	–39
	2	0.084	0.080	0.070	0.049	–42
	4	0.078	0.088	0.077	0.050	–35
integrated, $t = 5-10$	1	0.095	0.085	0.072	0.054	–43
	2	0.093	0.100	0.084	0.056	–40
	4	0.087	0.115	0.098	0.061	–30

and Re = 200 with free and no slip; the six rows that differ are those of the terms that vanish at $t = 0$ and print round-off there. The free-slip box equals its periodic twin the same way. The sixteen runs of the test reproduce round 2’s time series and profiles exactly, and in the verification job that preceded them the device’s budget agreed with the host’s to 1.6×10^{-15} of each term’s scale, with free and no slip. On the stretching, a check the physics gives for free: on the wall nodes of a no-slip conducting box S vanishes identically, since the velocity and its tangential derivatives vanish and the field is tangential.

Closure. Products of central differences do not telescope, and the compressible remainder oscillates with the acoustic waves at the size of a layer’s Ohmic dissipation, so the explicit terms close a layer’s budget only to a residual. Over $t = 2-10$, on the finer grids, the magnetic budget within δ closes to 1.0–2.9% of the Ohmic dissipation there in A and 0.3–0.7% in B. The kinetic one closes to 0.1–0.7% of the viscous dissipation in B and to 1.3–10.4%

in A, whose near-wall viscous dissipation is small. Over the whole box the magnetic and kinetic budgets close to 1.6% and 1.2% of the dissipation in every run, and $\langle S \rangle$ balances $-\langle \mathbf{v} \cdot (\mathbf{j} \times \mathbf{B}) \rangle$ to 0.9%. The residual falls at second order: in previews at Re = 200, doubling N from 65 to 129 took the magnetic closure within δ from 15% to 3.1% in A and from 11% to 1.7% in B. The test of Sec. IV G does not use the explicit transports, only S and the Ohmic dissipation.

What had been seen when the test was fixed. The previews were run to check these properties. One of them, at $N = 65$ to $t = 2$, had been analysed for its closure, in a table that showed S and the Ohmic dissipation within δ over $t \leq 2$; no preview over the test’s window had been. Read afterwards, the $N = 65$ preview at Re = 200, which is not a rung and has 1.4 cells in δ , gives $\phi = 2.5$.

TABLE S4. The test of Sec. IV G, rung by rung, over $t = 2$ –10 within δ : ϕ on the finer and the coarser grid, its band, the transport and storage shares of the Ohmic deficit (the three adding to one), the near-wall stretching of each box as a share of its own near-wall Ohmic dissipation, and the magnetic budget’s closure within δ in A and B on the finer grid.

Re	N	ϕ	ϕ_{coarse}	band	transport	storage	S_A/Ω_A	S_B/Ω_B	closure (%)
250	192, 256	1.86	1.89	0.035	+0.06	−0.92	0.57	−0.40	1.0, 0.7
500	256, 384	1.55	1.58	0.037	−0.10	−0.44	0.56	−0.26	1.2, 0.3
1000	384, 512	1.47	1.51	0.035	−0.30	−0.17	0.56	−0.08	1.9, 0.4
2000	512, 640	1.84	1.89	0.052	−0.91	+0.07	0.67	−0.22	2.9, 0.5

S8. THE CONTROLS

The four controls of Sec. IV H were fixed, with their rules, in the job script that ran them, committed and pushed before the runs (Table S1); Table S5 gives every number. All 22 runs reached $t = 10$.

The start. B with its walls moving with the flow at $t = 0$ and brought to rest along half a cosine over $T = 0.2$ or 1.0, at Re = 500 ($N = 384$) and 2000 ($N = 640$), against round 2’s A on the same grid. The rule allowed B–A to change by less than 10% of its value, about five times its band at these rungs, and required it to stay above its band. A smooth start needs a wall velocity that differs at every wall node, which the regularized wall’s table of wall states could not hold; both implementations now store one per node, and every earlier reference is reproduced byte for byte. At $t = 0$ a smooth start is A exactly.

The field at matched energy. C with an initial velocity amplitude of $\sqrt{2}v_0$, so $E_V(0) = \frac{1}{4}$, on both grids of every rung. The rule required F_1 of C to exceed F_1 of B by more than both bands at every rung. This C has a turbulent peak at Re = 500, 1000 and 2000 ($t = 2.10, 2.39$ and 7.10) and none at 250.

The Mach number. A and B at Re = 2000, $N = 640$, $u_0 = 0.04$, where $\tau - \frac{1}{2} = 0.0122$ sits on the regularized wall’s stability floor, as the Re = 1000 Mach run did; halving u_0 would need $N \approx 1024$, which one GPU cannot hold. The largest lattice Mach number falls from 0.133 to 0.107.

The magnetic Prandtl number. A and B at Re = 1000 with Pm = 0.5 on $N = 384$ and 512, and with Pm = 2

on $N = 512$ and 640 in single precision, like the Re = 2000 rung, whose magnetic relaxation parameters these reproduce. The bands of B–A are 0.0003 and 0.0008, and of ϕ 0.017 and 0.031.

On the device. The per-node wall velocity has a half that only the GPU compiler sees, so the verification job that preceded these runs ran a smooth start and the energy-matched box on the device against host references: they agree to 5.5×10^{-16} of each diagnostic’s scale, and their budgets to 1.7×10^{-15} .

S9. THE FIELDS AT A LATER TIME

Figures S1 and S2 repeat Figs. 4 and 5 at $t = 4.6$, by the later maxima of the dissipation. By then the bulk of both MHD boxes is more finely structured and A’s near-wall plane has quietened, but the contrast of Sec. IV F remains: in the plane 0.58δ above the bottom wall, B’s viscous dissipation is 6.8 times A’s and its Ohmic 0.68 of A’s. In three dimensions both MHD boxes have broken up into finer sheets, and the viscous dissipation of the no-slip box forms ring-like patterns on its faces. The box without a field dissipates 3.3 times its box mean in its near-wall plane, about 3.2 times B’s relative level.

¹A. Pouquet, E. Lee, M. E. Brachet, P. D. Mininni, and D. Rosenberg, “The dynamics of unforced turbulence at high Reynolds number for Taylor–Green vortices generalized to MHD,” *Geophys. Astrophys. Fluid Dyn.* **104**, 115–134 (2010).

²J. Sommeria and R. Moreau, “Why, how, and when, MHD turbulence becomes two-dimensional,” *J. Fluid Mech.* **118**, 507–518 (1982).

³B. Knaepen and R. Moreau, “Magnetohydrodynamic turbulence at low magnetic Reynolds number,” *Annu. Rev. Fluid Mech.* **40**, 25–45 (2008).

TABLE S5. The four controls of Sec. IV H, over $t = 2-10$ within δ of the walls, on the finer grid: each quantity in the reference run (round 2, impulsive start, $u_0 = 0.05$, $\text{Pm} = 1$) and in the control. For C2 the two values are F_1 of B and F_1 of C at B's initial energy.

control	Re	change	quantity	reference	control	shift
C1, start	500	ramp 0.2	B-A	0.0551	0.0579	+5.0%
C1, start	500	ramp 1.0	B-A	0.0551	0.0663	+20.3%
C1, start	2000	ramp 0.2	B-A	0.0481	0.0480	-0.2%
C1, start	2000	ramp 1.0	B-A	0.0481	0.0491	+2.2%
C2, field	250	C at $E_V(0) = \frac{1}{4}$	F_1 of B, of C	0.1278	0.2889	
C2, field	500	C at $E_V(0) = \frac{1}{4}$	F_1 of B, of C	0.1108	0.2770	
C2, field	1000	C at $E_V(0) = \frac{1}{4}$	F_1 of B, of C	0.0994	0.2613	
C2, field	2000	C at $E_V(0) = \frac{1}{4}$	F_1 of B, of C	0.0861	0.2083	
C3, Mach	2000	$u_0 = 0.04$	B-A	0.0481	0.0480	-0.00007
C3, Mach	2000	$u_0 = 0.04$	ϕ	1.839	1.835	-0.003
C4, Pm	1000	$\text{Pm} = 0.5$	B-A	0.0539	0.0597	
C4, Pm	1000	$\text{Pm} = 0.5$	ϕ	1.474	2.010	
C4, Pm	1000	$\text{Pm} = 2$	B-A	0.0539	0.0519	
C4, Pm	1000	$\text{Pm} = 2$	ϕ	1.474	1.221	

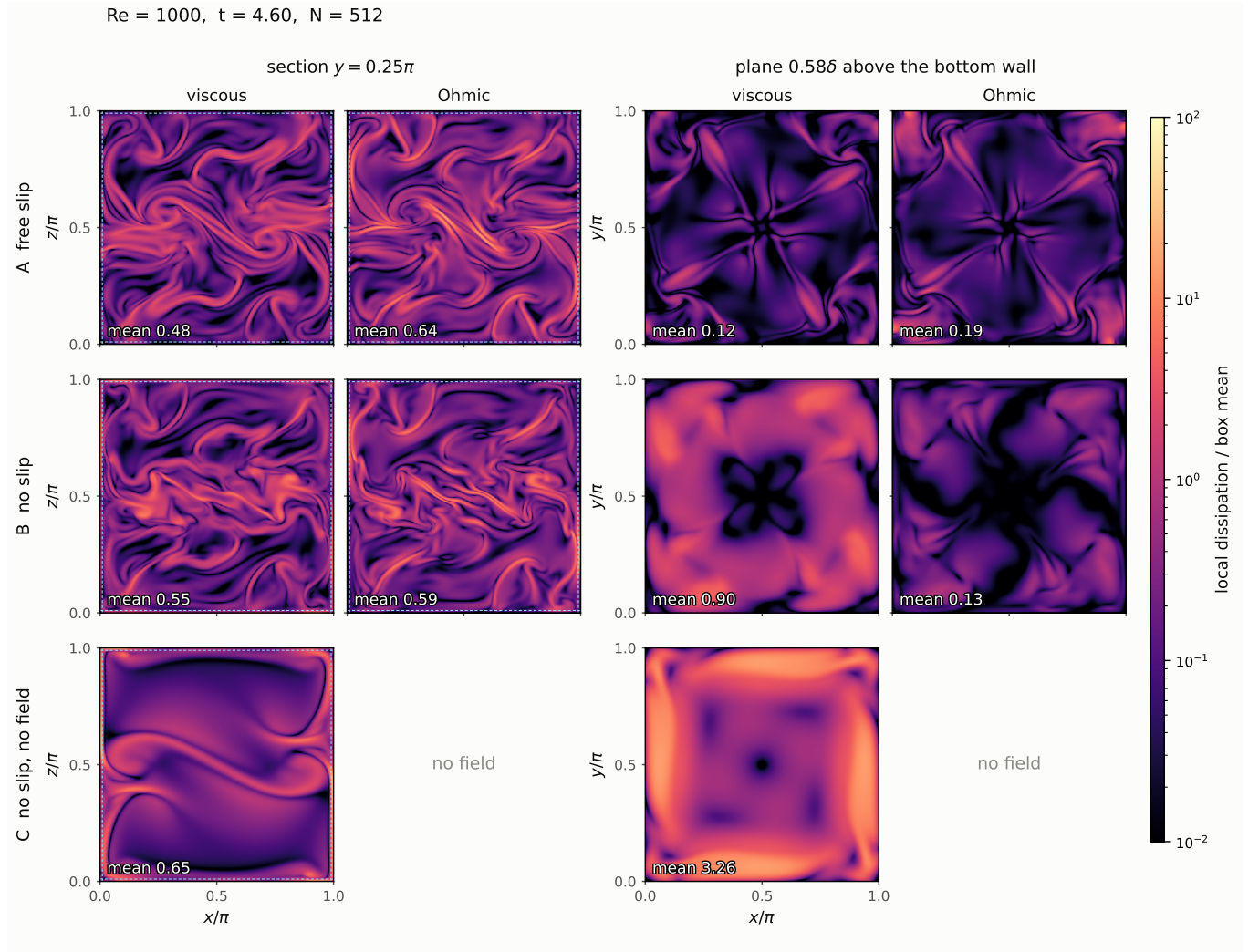


FIG. S1. As Fig. 4, at $t = 4.6$, by the later maxima of the dissipation.

$Re = 1000$, $t = 4.60$, $N = 512$ (block maximum over 2^3)

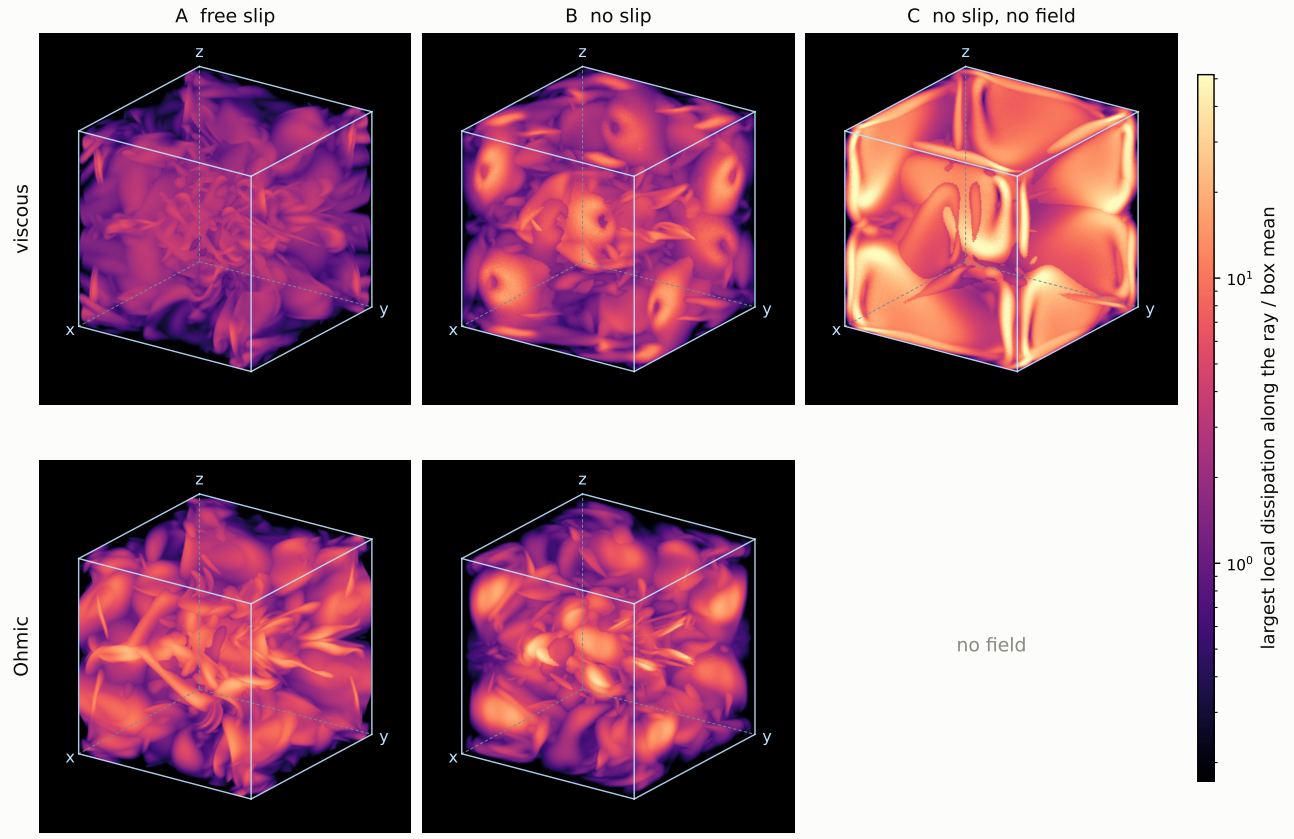


FIG. S2. As Fig. 5, at $t = 4.6$, on the same colour scale.